BLOCKCHAIN-BASED DIGITAL RECEIPT AND PURCHASE HISTORY MANAGEMENT SYSTEM

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Use Case Report

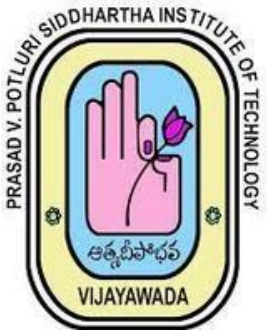
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CERTIFICATE

This is to certify that the Use Case report entitled **"Blockchain-Based Digital Receipt and Purchase History Management System"** that is being submitted by **Darsi Himaja (23501A0533)**, as part of Assignment-2 for the **Blockchain Technology (23CS4701B)** course in **4-1** during the academic year **2026-27**.

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MARKS

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain is a distributed ledger in which records are grouped into blocks and linked together using cryptographic hashes. Every participating node keeps a copy of the ledger, and a new block is added only after the network agrees that its contents are valid, using a consensus mechanism. Because each block carries the hash of the block before it, changing one old record would break every block that follows, which makes silent tampering practically impossible. Although blockchain first became popular as the foundation of cryptocurrencies such as Bitcoin, its ability to provide a shared and trustworthy record without a central authority has led to its use in finance, supply chains, healthcare, education, and retail.

1.2. Relevance of Blockchain in Digital Receipt Systems

In everyday retail, a receipt is the main proof that a purchase took place. It is needed for returns and exchanges, warranty claims, expense reimbursement, insurance claims, and tax filing. Yet most receipts are still printed on thermal paper that fades within months, or are emailed in formats that end up scattered across inboxes and shop-specific apps. Each retailer keeps its own records in its own database, so a customer has no single place to see the complete purchase history and no independent way to prove that a record has not been altered.

A blockchain-based digital receipt system addresses these gaps by issuing every receipt as a tamper-proof entry on a shared ledger. When a merchant completes a sale, a smart contract records the merchant, the buyer, the amount, the time of purchase, and a hash of the purchased items against the buyer's blockchain address. The receipt then stays permanently linked to that address, so the buyer can view the entire purchase history from one place, and anyone who needs to check a receipt, such as a service centre or an auditor, can verify it directly on the chain. Proposals such as ERC-5570 already describe how receipts for physical purchases can be represented as tokens whose metadata carries the receipt details.

This approach builds trust between merchants and customers because neither side can edit or deny a record once it has been written. Smart contracts remove manual steps from issuing and verifying receipts, which reduces errors and the time spent on disputes. Academic prototypes such as the BlockReceipt project at the University of Hong Kong also show that a paperless, fraud-resistant alternative to conventional receipts is practical.

Since the ledger is decentralized, receipt data is available from anywhere and does not disappear when a single shop closes, changes its software, or loses its database. Customers can keep a lifelong record of their purchases, which is valuable for warranty, insurance, and personal budgeting. For merchants, a verifiable receipt trail simplifies audits and the handling of returns. As more businesses move to digital payments, a blockchain-based receipt system offers a scalable and future-ready way of managing proof of purchase securely and efficiently.

2. BACKGROUND

Conventional receipt and purchase-record management depends on paper slips, shop-specific databases, and scattered email copies. These methods are convenient at the point of sale, but they create many problems afterwards, when the receipt is actually needed. Below are the key issues with existing systems:

2.1. Loss, Fading and Damage of Paper Receipts

Most paper receipts are printed on thermal paper, which fades with heat, light, and time, and is easily lost or crumpled. When a customer needs to return a product or make a warranty claim months later, a faded or missing slip often leads to the claim being refused, even though the purchase was genuine.

2.2. Forged and Duplicate Receipts

Paper and PDF receipts can be photocopied, edited, or fabricated with simple tools. They are then used for fraudulent returns, inflated expense claims, or duplicate reimbursements. Without a reliable way to verify authenticity, merchants and employers must either trust the document or spend effort checking it manually.

2.3. Fragmented Digital Receipts

Even electronic receipts are scattered across email accounts, SMS messages, and individual store apps, each with a different format. A customer cannot easily search or compile all purchases in one place, and a receipt held only in a retailer's own system may be lost if the retailer changes its platform or closes down.

2.4. Lack of Transparency and Independent Verification

Purchase records sit in centralized databases controlled by the seller. Customers cannot independently verify what is stored, and third parties such as service centres, banks, or tax officers have to rely on the seller's confirmation. This leads to disputes over whether a purchase happened, when it happened, and how much was paid.

2.5. Difficulties in Returns, Exchanges and Warranty Claims

Proof of purchase is usually checked by hand at the service counter, which is slow and inconsistent between stores. There is also no common way to check whether an item has already been returned or refunded elsewhere, so the same purchase can be claimed more than once.

2.6. Security Risks and Data Tampering

Centralized databases are attractive targets for cyberattacks and are also exposed to insider modification. When a record is changed in a central system, it is hard for the customer to detect the change. A breach can also leak the buying habits and personal details of thousands of customers at once.

2.7. Difficulty in Expense Tracking, Auditing and Tax Compliance

Individuals and businesses have to collect, scan, and store receipts for reimbursement, auditing, and tax purposes, often for several years. Manual collection is time-consuming and error-prone, and missing receipts can lead to rejected claims or compliance issues.

2.8. Disputes Between Merchants and Customers

When the customer's copy and the merchant's record do not match, there is no neutral source of truth to settle the matter. Disagreements about price, date, or items purchased are therefore resolved on the basis of whoever can present more convincing paperwork, which often damages the customer's trust in the brand.

2.9. Limited Consumer Access and Control over Purchase Data

Customers rarely have easy access to their full purchase history from a retailer. They usually need to contact customer support or log in to separate shop accounts, and they have little control over who can see or use their purchase data. A single, customer-owned view of all purchases is not available in the traditional model.

2.10. Environmental Impact of Paper Receipts

A very large number of paper receipts are printed every day, and most are thrown away within minutes of the purchase. Thermal paper carries a chemical coating that generally prevents it from being recycled with ordinary paper, so reliance on paper receipts adds to avoidable waste even as businesses and consumers move towards digital payments.

Because of these challenges, there is a clear need for a secure, transparent, and automated way of issuing and storing receipts. Blockchain technology offers a promising solution by recording each receipt as a tamper-proof entry, giving customers a unified purchase history, and allowing any authorized party to verify a receipt without depending on a single central database.

3. BLOCKCHAIN BASICS

Blockchain is a distributed ledger system that provides secure, transparent, and tamper-resistant record keeping. Instead of relying on an intermediary to maintain and vouch for the records, a blockchain lets many independent participants share and verify the same data. This makes it well suited to applications such as payments, supply chain tracking, and digital receipt management. The following are the key concepts that define blockchain technology:

3.1. Decentralization

Traditional databases are managed by a central authority such as a bank or a company, which creates a single point of failure and makes them vulnerable to hacking and data manipulation. A blockchain instead runs on a network of independent nodes, each of which validates and stores the data. No single entity controls the system, which makes it resistant to censorship and fraud. Compromising one node does not affect the integrity of the network, and since every transaction is verified by many nodes, participants can independently check the records.

3.2. Immutability

Immutability means that once a transaction has been recorded on the blockchain, it cannot be altered or deleted. This is achieved through cryptographic hashing: each block stores the hash of the previous block, forming a chain. To change one old block, an attacker would have to recompute every later block and convince most of the network to accept the new version, which is extremely difficult. This property makes blockchain particularly valuable for audit trails, financial records, and proof of purchase.

3.3. Transparency

Blockchain transactions can be verified by any participant who has access to the network. With distributed ledger technology, every participant holds an identical copy of the data, so hidden alterations are not possible. Public blockchains such as Bitcoin and Ethereum are fully transparent, and permissioned blockchains can offer controlled transparency among approved parties. In either case, any attempt to manipulate records is quickly noticed by the network.

3.4. Smart Contracts

A smart contract is a program that runs on the blockchain and executes automatically when predefined conditions are met. It consists of code (its functions) and data (its state) stored at a specific address, and interactions with it are recorded as transactions. [4] Because the rules are enforced by code instead of by an intermediary, smart contracts reduce manual work, cost, and disputes. In a digital receipt system, smart contracts can register merchants, issue receipts, return a customer’s purchase history, verify the authenticity of a receipt, and record refunds without any human intervention.

3.5. Consensus Mechanisms

Consensus mechanisms allow all nodes to agree on which transactions are valid and in what order they are added. In Proof of Work (PoW), used by Bitcoin, miners compete to solve computational puzzles. In Proof of Stake (PoS), validators are selected based on the amount of cryptocurrency they lock up as a stake; Ethereum has used PoS since its 2022 transition. Variations such as Delegated Proof of Stake (DPoS) and Proof of Authority (PoA) use a smaller

set of validators to improve speed and reduce energy use. These mechanisms prevent malicious actors from inserting invalid data into the ledger.

3.6. Cryptographic Security

Blockchain security relies on public-key cryptography and hash functions. Every user has a private key that is used to sign transactions and a public key (address) that others use to verify them, so only the rightful owner of an address can act on its behalf. Hash functions convert data into fixed-length codes that are practically impossible to reverse, and even a tiny change in the input produces a completely different hash. In a receipt system, the hash of the purchased items stored with each receipt allows anyone to confirm that the receipt details have not been changed.

By combining decentralization, immutability, transparency, smart contracts, consensus, and cryptographic security, blockchain provides a robust foundation for managing digital receipts. It makes issuing, storing, and verifying proof of purchase secure, automated, and trustworthy for every party involved.

4. USE CASE OVERVIEW

The use case for a Blockchain-Based Digital Receipt and Purchase History Management System is to change the way proof of purchase is issued, stored, and verified. The system gives every purchase a secure, tamper-proof digital receipt, keeps all receipts of a customer in one place, and lets authorized parties verify them without depending on a central database.

4.1. Objectives

The primary objective of the system is to replace paper and scattered email receipts with a fully digital platform in which every receipt is stored on a blockchain. Receipts stored this way cannot fade, be lost, or be misplaced, and they are accessible whenever the customer needs them.

A second objective is to prevent forged and duplicate receipts. Because a receipt is written to the blockchain by a registered merchant and cannot be edited afterwards, it is not possible to create a fake receipt that passes verification. This protects merchants from fraudulent returns and protects employers and insurers from false claims.

Another important goal is to give customers a unified purchase history. All receipts issued to a customer's blockchain address can be retrieved with a single call, regardless of which registered merchant issued them. This removes the need to search through emails, apps, and paper bills when a receipt is required.

The system also aims to speed up returns, exchanges, and warranty claims. A service centre or returns desk can verify a receipt instantly by checking it on the blockchain, and the smart contract records when a receipt has been refunded so that the same purchase cannot be refunded twice.

Transparency and trust are further objectives. Merchants, customers, and verifiers all see the same tamper-proof record, which reduces disputes about what was bought, when, and for how much. By removing manual paperwork and verification effort, the system can also reduce operating costs for merchants, while the use of a decentralized ledger improves resistance to data loss and unauthorized modification.

4.2. Scope of the System

The blockchain-based receipt system focuses on securely issuing, storing, and verifying digital receipts and purchase history. The system includes:

1. **Merchants (Registered Retailers)**: Register on the platform and issue digital receipts for each sale, which are recorded on the blockchain.
2. **Customers (Buyers)**: View their complete purchase history, share a receipt as proof of purchase, and use it for returns, warranty, or reimbursement.
3. **Verifiers (Service Centres, Returns Desks, Auditors)**: Verify a receipt in real time without relying on the merchant's private database.
4. **Blockchain Network**: A decentralized ledger that stores receipts in a tamper-proof manner and executes the smart contract.

4.3. System Architecture

The architecture of the Blockchain-Based Digital Receipt and Purchase History Management System consists of the following layers:

4.3.1. User Layer

This layer consists of the primary users who interact with the system. (**Figure 4.1 - User Layer**).

- **Merchants**:
  - Register as a merchant and issue a digital receipt for every sale.
  - Mark a receipt as refunded when a return is accepted.
- **Customers**:
  - View all receipts linked to their blockchain address as a purchase history.
  - Present a receipt as proof of purchase for returns, warranty, or reimbursement.
- **Verifiers**:
  - Check that a receipt exists and that its details match the items hash on the blockchain.
  - Confirm that a receipt has not already been refunded before approving a claim.

4.3.2. Web Application Layer

This layer acts as the interface between the users and the blockchain. (**Figure 4.1 - Web Application Layer**).

- **User Interface**:
  - Allows merchants to create receipts at the point of sale and manage refunds.
  - Enables customers to browse, search, and share their purchase history.
  - Provides verifiers with a simple screen to look up and verify a receipt.
- **Receipt Management System**:
  - Handles user registration and wallet-based authentication through MetaMask.
  - Facilitates communication between users and the blockchain using Web3.js or Ethers.js.

4.3.3. Blockchain Layer

This layer ensures the security, transparency, and immutability of receipt transactions. (**Figure 4.1 - Blockchain Layer**).

- **Smart Contract Layer**:
  - **DigitalReceipt Contract**: Stores receipt details such as merchant, buyer, amount, timestamp, and items hash.
  - Manages merchant registration, receipt issuing, purchase history retrieval, verification, and refund marking.
- **Smart Contract Execution**:
  - When a merchant issues a receipt, the contract records it on the blockchain and links it to the buyer's address.
  - When a return is accepted, the contract flags the receipt as refunded without deleting it, so the history stays intact.

- **Blockchain Network**:
  - **Distributed Ledger**: Maintains a decentralized, tamper-proof record of all receipts.
  - **Immutable Transactions**: Ensures receipt data cannot be modified, preventing fraud.

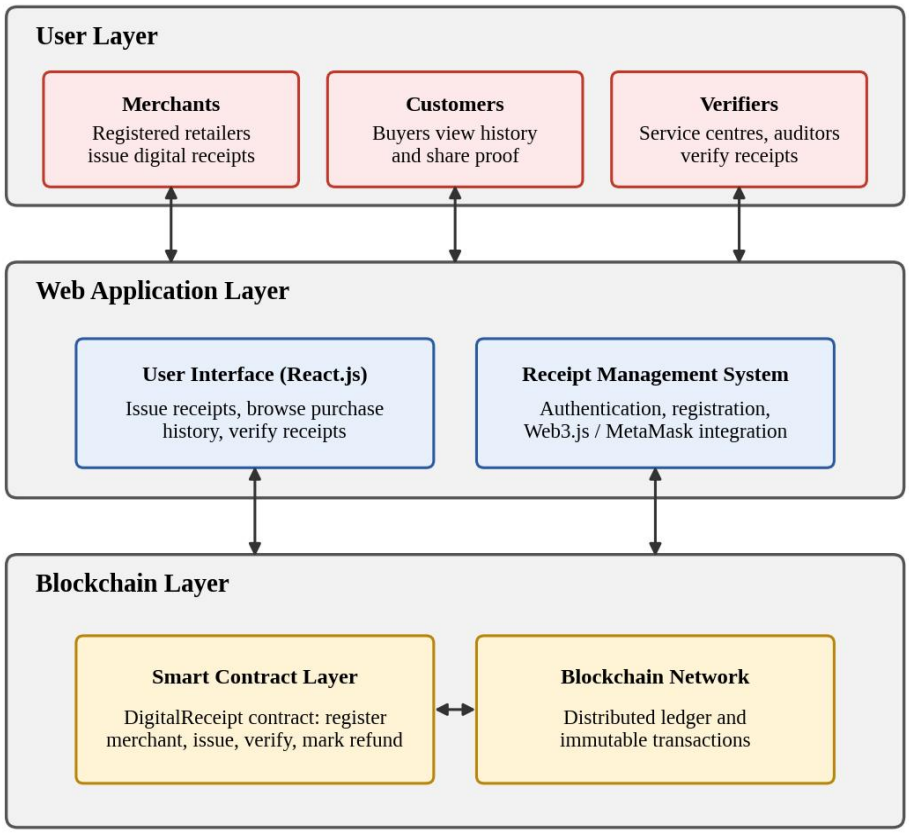


Figure 4.1: Architecture of Blockchain based Digital Receipt and Purchase History Management System

5. IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

To implement the system, the development environment is set up using Ethereum as the blockchain platform. Ganache provides a personal Ethereum blockchain on the local machine for fast testing, and a development framework such as Truffle (or Hardhat) is used to compile, deploy, and test the smart contract. MetaMask is integrated to manage blockchain accounts and sign transactions. The smart contract is written in Solidity (v0.8.19), which includes built-in overflow checks and clearer error handling. Once the environment is ready, the receipt management smart contract can be written.

5.2. Defining Smart Contracts

5.2.1. Writing the Smart Contract for Receipt Management

The DigitalReceipt contract is responsible for registering merchants, issuing receipts, returning purchase history, verifying receipts, and recording refunds. It defines a Receipt struct that holds the receipt ID, the merchant, the buyer, the amount, the time of issue, a hash of the purchased items, a short item description, and a refunded flag. The contract uses three mappings: one to track registered merchants, one to store receipts by ID, and one to associate each buyer address with the list of receipt IDs issued to it. This structure keeps every receipt securely linked to its rightful owner and makes the purchase history easy to retrieve.

```
struct Receipt {
    uint256 id;
    address merchant;
    address buyer;
    uint256 amount;
    uint256 issuedAt;
    bytes32 itemsHash;
    string  itemDetails;
    bool    refunded;
}

uint256 internal nextReceiptId = 1;
mapping(address => bool) internal registeredMerchants;
mapping(uint256 => Receipt) internal receipts;
mapping(address => uint256[]) internal buyerToReceipts;
```

5.2.2. Registering Merchants

To prevent unauthorized entities from issuing receipts, a merchant must register its blockchain address before it can generate receipts. This is enforced by a function that allows only new merchants to register. The function ensures that a merchant cannot register more than once and helps maintain a verified list of trusted retailers. By restricting receipt issuing to registered merchants, the system prevents the creation of fake receipts.

```
function registerMerchant() external {
    require(!registeredMerchants[msg.sender], "Already registered");
    registeredMerchants[msg.sender] = true;
    emit MerchantRegistered(msg.sender);
}
```

5.2.3. Issuing Digital Receipts

Once registered, a merchant can issue a receipt for every sale. The merchant provides the buyer's blockchain address, the amount paid, and the item details. The contract assigns a unique receipt ID, records the time of issue from the block timestamp, computes a keccak256 hash of the item details, and stores the receipt under the buyer's address. An event is emitted so that the front end can update immediately. Since the receipt is written to the blockchain, it is immutable, tamper-proof, and permanently available to the buyer.

```
function issueReceipt(address buyer, uint256 amount,
                      string memory itemDetails)
    external returns (uint256)
{
    require(registeredMerchants[msg.sender], "Not a merchant");
    require(buyer != address(0), "Invalid buyer address");
    require(amount > 0, "Invalid amount");

    uint256 receiptId = nextReceiptId++;
    receipts[receiptId] = Receipt({
        id: receiptId,
        merchant: msg.sender,
        buyer: buyer,
        amount: amount,
        issuedAt: block.timestamp,
        itemsHash: keccak256(bytes(itemDetails)),
        itemDetails: itemDetails,
        refunded: false
    });

    buyerToReceipts[buyer].push(receiptId);
    emit ReceiptIssued(receiptId, msg.sender, buyer, amount);
    return receiptId;
}
```

5.2.4. Viewing the Purchase History of a Customer

Customers must be able to access and review all of their receipts. This is achieved through a read-only function that retrieves every receipt linked to the caller's blockchain address. The function loops through the stored receipt IDs and returns a structured list containing the merchant, amount, date, and item details of each purchase. Because it is a view function, it does not cost gas when called from the front end, and it gives customers a complete and transparent purchase history.

```
function getPurchaseHistory()
    external view returns (Receipt[] memory)
{
    uint256[] memory ids = buyerToReceipts[msg.sender];
    Receipt[] memory history = new Receipt[](ids.length);

    for (uint256 i = 0; i < ids.length; i++) {
        history[i] = receipts[ids[i]];
    }
    return history;
}
```

5.2.5. Verifying Receipt Validity

Before approving a return or a warranty request, a service centre or returns desk must verify that the receipt is genuine. The verifyReceipt function checks that a receipt with the given ID exists, that the items hash supplied matches the hash stored on the blockchain, and that the receipt has not already been refunded. It returns true only when all three conditions hold. This prevents forged receipts and duplicate refund claims, and ensures that only legitimate purchases receive after-sales service.

```
function verifyReceipt(uint256 receiptId, bytes32 itemsHash)
    external view returns (bool)
{
    Receipt storage r = receipts[receiptId];
    return r.id != 0 && r.itemsHash == itemsHash && !r.refunded;
}
```

5.2.6. Handling Returns and Refunds

A crucial feature of the system is the correct handling of returns. When a return is accepted, the issuing merchant calls the markRefunded function. The function ensures that only the merchant that issued the receipt can perform this action and that the receipt has not been refunded before. Instead of deleting the receipt, the contract sets its refunded flag, so the complete purchase history stays on record while the receipt can no longer be used for another claim. This keeps the ledger honest and auditable at all times.

```
function markRefunded(uint256 receiptId) external {
    Receipt storage r = receipts[receiptId];
    require(r.id != 0, "Receipt not found");
    require(r.merchant == msg.sender, "Only issuing merchant");
    require(!r.refunded, "Already refunded");

    r.refunded = true;
    emit ReceiptRefunded(receiptId);
}
```

5.3. Deploying and Integrating with a Frontend

After the smart contract has been developed and tested on Ganache, it is deployed to an Ethereum test network such as Sepolia before any move to a production network or a Layer 2 network. The front end is developed using React.js, with Web3.js or Ethers.js connected to the deployed contract through MetaMask. Merchants can register, issue receipts, and mark refunds, customers can browse their purchase history, and verifiers can look up a receipt by its ID, all through a simple interface that does not require any technical blockchain knowledge.

5.4. Workflow

The workflow of the Blockchain-Based Digital Receipt and Purchase History Management System, as illustrated in **Figure 5.1**, describes the interactions between merchants, customers, and verifiers for the secure issuing, viewing, and verification of receipts.

The process begins with the merchant registering on the platform. When a customer buys a product, the merchant issues a digital receipt through the smart contract, and the receipt is stored

on the blockchain and linked to the buyer's address. The customer can then view the receipt in a unified purchase history at any time.

If a return, refund, or warranty claim is needed, the customer presents the receipt to a verifier. As per **Figure 5.1**, the verifier looks up the receipt ID and compares the items hash with the one stored on the blockchain. If the receipt is valid and has not already been refunded, the claim is processed and the receipt is marked as refunded. If the receipt is invalid or has already been refunded, the claim is rejected. If no claim is needed, the receipt simply stays in the customer's purchase history.

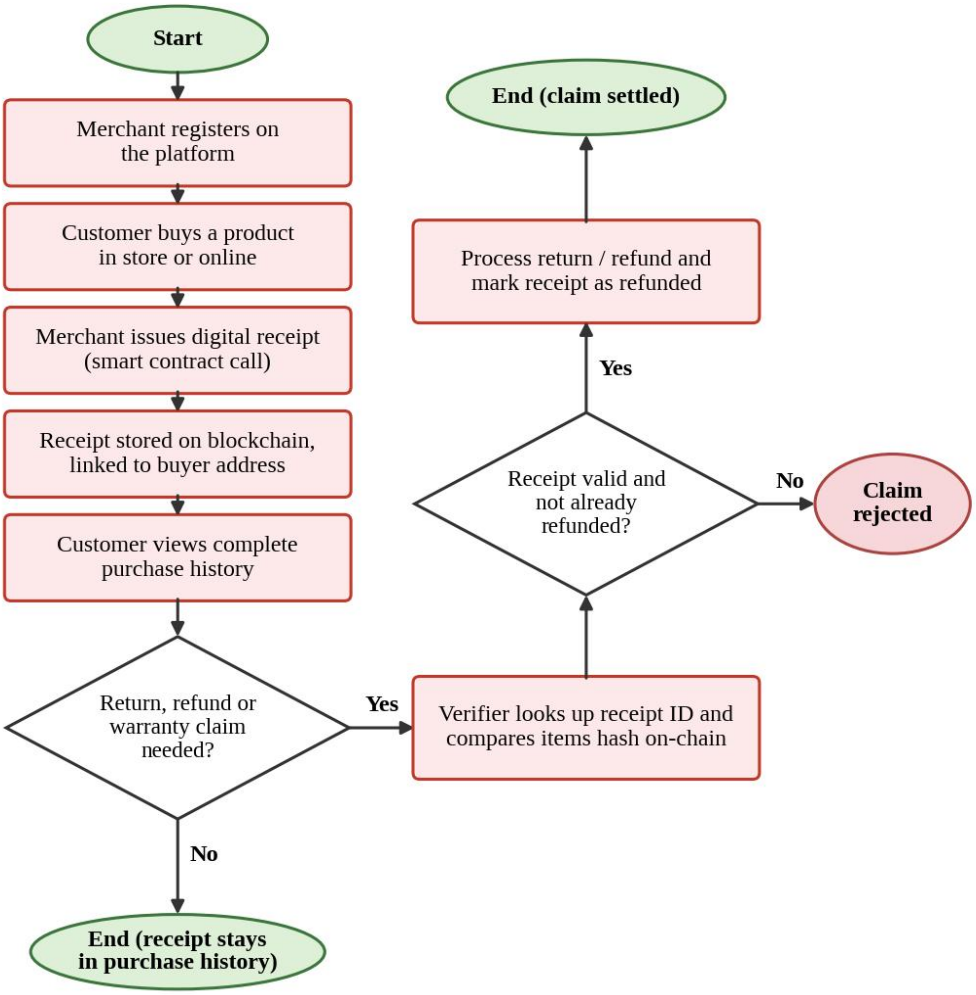


Figure 5.1: Workflow of Blockchain based Digital Receipt and Purchase History Management System

6. BENEFITS

Blockchain technology improves receipt management by adding transparency, security, and automation. Traditional receipts depend on paper and centralized databases that are prone to loss, forgery, and disputes. By using blockchain, receipt and purchase history management becomes tamper-proof, efficient, and easy to verify. The key advantages are:

6.1. Enhanced Security and Immutability

Once a receipt is written to the blockchain, it cannot be changed or deleted. This ensures that purchase records stay tamper-proof, and the cryptographic nature of the blockchain protects sensitive data from unauthorized modification.

6.2. Elimination of Forged and Duplicate Receipts

Every receipt is issued by a registered merchant and stored as a unique, verifiable record. A forged or edited receipt will not match the hash on the chain, so it fails verification. This prevents fraudulent returns and false expense claims and increases trust among merchants, verifiers, and customers.

6.3. Unified Purchase History

All receipts issued to a customer's address can be retrieved together, regardless of the shop, the date, or the payment method. Customers no longer need to search through emails, apps, and paper bills to find a single receipt, and the history is available for as long as the blockchain exists.

6.4. Transparency and Auditability

All parties see the same record, and every transaction is time-stamped on a distributed ledger. Auditors and tax officers can verify receipts without depending on intermediaries, which reduces disputes and increases customer confidence in the system.

6.5. Faster Returns and Warranty Claims

A returns desk or service centre can verify a receipt in seconds by checking it on the blockchain, instead of examining paper by hand or calling the merchant. This reduces queues and gives customers a quicker resolution of their claims.

6.6. Cost Reduction and Efficiency

The system reduces the cost of printing, storing, and manually verifying receipts. As all records are stored on-chain, merchants need less paperwork and fewer administrative staff for returns handling, and the resulting processes are faster and cheaper to run.

6.7. Smart Contracts for Automated Processing

Smart contracts automate the registration of merchants, issuing of receipts, verification, and refund marking. When a claim is made, the contract checks whether the receipt exists, matches the items hash, and has not already been refunded, which reduces processing time and human error.

6.8. Prevention of Duplicate Refunds

Because a refunded receipt is flagged on the ledger, it cannot be used again, even at a different branch of the same merchant. This closes a common loophole in traditional systems, where the same paper receipt may be presented repeatedly.

6.9. Increased Customer Trust and Satisfaction

Customers who can easily verify their purchases, access their receipts at any time, and rely on tamper-proof records are more likely to trust a merchant. This transparency can lead to higher customer retention and loyalty.

6.10. Sustainability and Paperless Operation

A digital receipt system removes the need for printed receipts and thermal paper, reducing waste and the environmental impact of retail. Moving to a fully paperless process makes receipt management more sustainable and eco-friendly.

Implementing blockchain in receipt management brings security, efficiency, and transparency. It prevents fraud, reduces administrative costs, and improves the customer experience by making receipts verifiable, permanent, and easy to access, which makes it a future-proof solution for modern retail.

7. CHALLENGES

While a blockchain-based receipt system offers many advantages, there are challenges and limitations that must be considered for effective implementation. Some of the key challenges are:

7.1. Scalability and Transaction Costs

Public blockchains can process only a limited number of transactions per second, and every receipt written to the chain costs gas. In a busy retail environment with thousands of sales per hour, this can lead to delays and high fees. Storing long item descriptions directly on-chain increases the cost further.

7.2. Privacy of Purchase Data

Data on a public blockchain is visible to everyone, and purchase history can reveal personal habits and preferences. Even though addresses are pseudonymous, patterns can be analyzed to identify users. Moreover, data protection laws such as India's Digital Personal Data Protection Act, 2023 give individuals rights over their personal data, which is difficult to reconcile with the permanent nature of blockchain records.

7.3. Regulatory and Legal Compliance

Blockchain regulations differ between countries, and tax, invoicing, and consumer-protection rules apply to receipts. It can be challenging for businesses to make sure that a blockchain receipt satisfies the legal requirements of a valid tax invoice or proof of purchase in every region.

7.4. Lack of User Awareness and Wallet Management

Many merchants and customers are not familiar with blockchain wallets, private keys, and transaction fees. A customer who loses the private key of their wallet loses access to their purchase history, and educating users to manage keys safely remains a major adoption challenge.

7.5. Integration with Existing Systems and Connectivity

Retailers already use point-of-sale, billing, and accounting software, and connecting these systems to a blockchain requires additional development effort. The system also depends on internet connectivity for issuing and verifying receipts, so areas with poor connectivity may find it difficult to use.

Despite these challenges, blockchain-based receipt management offers transformative potential in reducing fraud, improving transparency, and simplifying proof of purchase. To overcome these limitations, businesses must focus on privacy-preserving design, regulatory compliance, user education, security audits, and scalable blockchain solutions.

8. CONCLUSION

This report has explored how blockchain technology can be applied to digital receipts and purchase history management, and it has highlighted the weaknesses of paper and scattered digital receipts: fading and loss, forged or duplicate receipts, lack of a unified history, and slow, manual verification. By using the immutability and decentralization of blockchain, the proposed system improves the security and trustworthiness of proof of purchase. Smart contracts automate the registration of merchants, the issuing of receipts, verification, and refund marking, which reduces dependence on intermediaries and gives users a smoother experience.

The implementation described in the report provides a structured approach to managing receipts securely. Registered merchants issue receipts that are linked to the buyer's address, customers can view their complete purchase history, and verifiers can confirm the authenticity of a receipt in real time. Refunded receipts are flagged and cannot be reused, so the system prevents duplicate claims while keeping the history intact. This transition to a blockchain-powered model supports the goals of reducing paperwork, improving customer satisfaction, and promoting sustainability.

8.1. Future Outlook for Enhancements

To improve the system further, the full receipt details can be kept in off-chain storage such as IPFS, with only the hash stored on the blockchain. This reduces gas cost and protects customer privacy while keeping the receipt verifiable. Layer 2 networks can be used to handle large transaction volumes quickly and cheaply, and a hybrid model that combines private and public chains could balance cost, privacy, and security.

Decentralized identity (DID) solutions can give customers more control over who can read their purchase history, with options to share a receipt selectively for a warranty claim or an expense report. QR code integration at the point of sale would allow a customer to receive a receipt by scanning a code, without needing to share any personal details with the shop. Following standards such as ERC-5570 would also make receipts interoperable with other applications and wallets.

Artificial intelligence can be added to analyze purchase history for spending insights, budgeting suggestions, and unusual refund patterns that point to fraud. Integration with warranty and insurance systems would allow a single receipt to trigger a warranty record automatically. By continuously improving the system with these advancements, blockchain-based receipt management can evolve into a secure, efficient, and widely adopted solution in retail.

9. SDG's ADDRESSED

The Blockchain-Based Digital Receipt and Purchase History Management System aligns with United Nations Sustainable Development Goal 12 (Responsible Consumption and Production) by replacing paper receipts with tamper-proof digital receipts, which reduces paper waste and gives customers a permanent, verifiable record of their purchases.

SDG 12: Responsible Consumption and Production

Replacing paper receipts with digital blockchain receipts reduces paper use and the chemical waste from thermal printing. A very large number of paper receipts are printed every day and thrown away within minutes. Thermal paper generally cannot be recycled with ordinary paper, so moving to digital receipts directly reduces avoidable waste.

The system also supports responsible consumption. Customers get a permanent, verifiable record of every purchase, so they can track returns, refunds, warranties, and spending without losing proof of purchase. This avoids unnecessary replacement of products caused by lost or faded receipts. For merchants, a tamper-proof receipt trail reduces fraudulent returns and duplicate refunds, which supports efficient and sustainable business practices.

By integrating blockchain, the Digital Receipt and Purchase History Management System aligns with SDG 12, promoting a paperless, transparent, and sustainable way of managing proof of purchase.

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11. APPENDIX A

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